

Strategic Marketing for Innovation & Technology Businesses (SMIT)

Creating Markets, Commercializing Innovation & Scaling Growth

Course Proposal for IIM Calcutta, 2026-27

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Note: Updated and expanded edition for the 2026–27 PGP, building on the course offered at IIM Calcutta between 2011 and 2020.

Background

Technology has become one of the primary drivers of competitive advantage across industries. Markets today are characterized by rapid innovation, continual product evolution, changing customer expectations, and the blurring of traditional industry boundaries. Products increasingly evolve into services, services into platforms, and organizations continually rethink how value is created, delivered, and captured. Successive waves of technological change continue to reshape customer experiences, business models, competitive dynamics, and industry structures, creating both significant uncertainty and unprecedented opportunities for innovation and growth.

Success in such environments is no longer determined solely by superior technology. It increasingly depends on an organization's ability to understand market opportunities, anticipate customer needs, commercialize innovation, design compelling business models, orchestrate partnerships, and continually adapt to changing competitive dynamics. These shifts have fundamental implications for marketing, product management, business strategy, and organizational capability.

Strategic Marketing for Innovation & Technology Businesses (SMIT) examines how organizations create, commercialize, and scale technology driven businesses. The course builds on the fundamental principles of marketing while integrating perspectives from technology management, product strategy, and business model innovation. Rather than focusing on any single technology or industry, the course develops enduring frameworks for understanding how technology evolution, market dynamics, and industry transformation interact to create competitive advantage.

The course combines enduring marketing and strategy frameworks with discussions around contemporary developments including artificial intelligence, enterprise software, cloud platforms, semiconductors, digital commerce, platform ecosystems, and other emerging technologies. Through a combination of classic

cases, simulations, contemporary illustrations, and classroom discussion, participants develop an integrated understanding of how technology markets evolve, how new markets are created, how innovations diffuse, and how organizations successfully commercialize and scale technology products, platforms, services, and solutions.

The course is intended for students aspiring to careers in product management, marketing, consulting, business development, entrepreneurship, strategy, venture investing, and general management within technology intensive industries. It is equally relevant for managers in traditional sectors who increasingly confront digital transformation, technology enabled business models, and innovation driven competition within their own industries.

Learning Objectives

Upon successful completion of the course, students will be able to:

1. **Develop** an integrated understanding of the distinctive characteristics of technology intensive markets and their implications for marketing, product strategy, business models, and competitive advantage.
2. **Analyse** how technology evolution, market dynamics, and industry transformation interact to create opportunities for innovation, disruption, and business growth.
3. **Design** integrated market strategies for creating, commercializing, and scaling technology products, platforms, services, and business models by adapting the principles of marketing to technology intensive environments.
4. **Evaluate** strategic alternatives for technology businesses by integrating customer understanding, product management, business model design, partnerships, ecosystem thinking, and commercialization decisions into coherent market strategies.

5. **Apply** enduring marketing and strategic frameworks to evaluate emerging technologies, identify new market opportunities, and build sustainable competitive advantage across successive waves of technological change.

Course Overview

Technology markets evolve rapidly, yet the fundamental principles underlying successful innovation, market creation, commercialization, and competitive advantage remain remarkably enduring. **Strategic Marketing for Innovation & Technology Businesses (SMIT)** goes beyond understanding today's technologies to develop the integrated marketing and strategic perspectives required to create, commercialize, and scale technology driven businesses across successive waves of technological change.

The course follows a structured learning journey organized into five interconnected blocks, each building on the previous one. Participants first develop an understanding of how technologies evolve and reshape markets, industries, and competitive dynamics. Building on this foundation, they examine how organizations identify market opportunities, design successful technology products, drive customer adoption, and create new markets. The course then explores how technology businesses are built through product strategy, business model innovation, partnerships, and platform thinking. It subsequently examines the challenges of scaling innovation, marketing technology offerings, and evolving from selling products to delivering integrated customer solutions. The final block integrates these perspectives by applying the course frameworks to evaluate emerging technologies and identify future business opportunities.

Throughout the course, enduring marketing and strategic frameworks are developed through the study of seminal cases, simulations, and landmark business initiatives, and are continually reinforced through **Technology Updates (Current Developments)** that connect classroom discussions with current and emerging

developments in technology markets. Rather than treating these developments as isolated topics, the course uses them to demonstrate how enduring principles continue to guide managerial decision making amidst continual technological change.

Designed as an elective delivered over **five two-day modules**, the course combines classic cases, simulations, structured discussions, contemporary illustrations, and integrative exercises. By the end of the course, participants will have developed an integrated perspective spanning **technology evolution, market development, and industry transformation**, together with a set of enduring marketing and strategic frameworks that can be applied to both today's technology markets and the next generation of technology businesses.

Teaching and Learning Approach

The course adopts a highly interactive, discussion-oriented approach that combines conceptual frameworks with contemporary business practice. Learning is built around a blend of **case discussions, simulations, classroom exercises, structured debates, short lectures, Technology Updates, expert perspectives, and curated readings** drawn from both academic research and business practice. Contemporary examples are continually integrated to connect enduring principles with the rapidly evolving technology landscape.

A central philosophy of the course is to encourage participants to **discover managerial insights through discussion before connecting them to established frameworks**. Cases, simulations, and classroom exercises are therefore designed to help participants analyse situations, debate alternative viewpoints, and develop their own perspectives before introducing the underlying conceptual models. Interactive tools such as live polling, collaborative exercises, and digital engagement platforms will be used where appropriate to encourage participation and facilitate real-time learning.

The course pack contains a carefully curated selection of required readings, case material, simulations, and supporting articles that serve as preparation for classroom discussions. Required readings for each session are outlined in the Session Plan, while additional references are provided for participants wishing to explore specific topics in greater depth. Class discussions are supplemented by contemporary examples, instructor notes, and additional material shared during the course.

To encourage continued learning beyond the classroom, participants are also provided with an extensive collection of recommended resources, including articles from academic and business journals, industry reports, podcasts, videos, TED Talks, interviews, and selected documentaries relevant to the session themes. This resource base has evolved over several years through successive offerings of the course and is continually updated to reflect emerging developments in technology, innovation, and business.

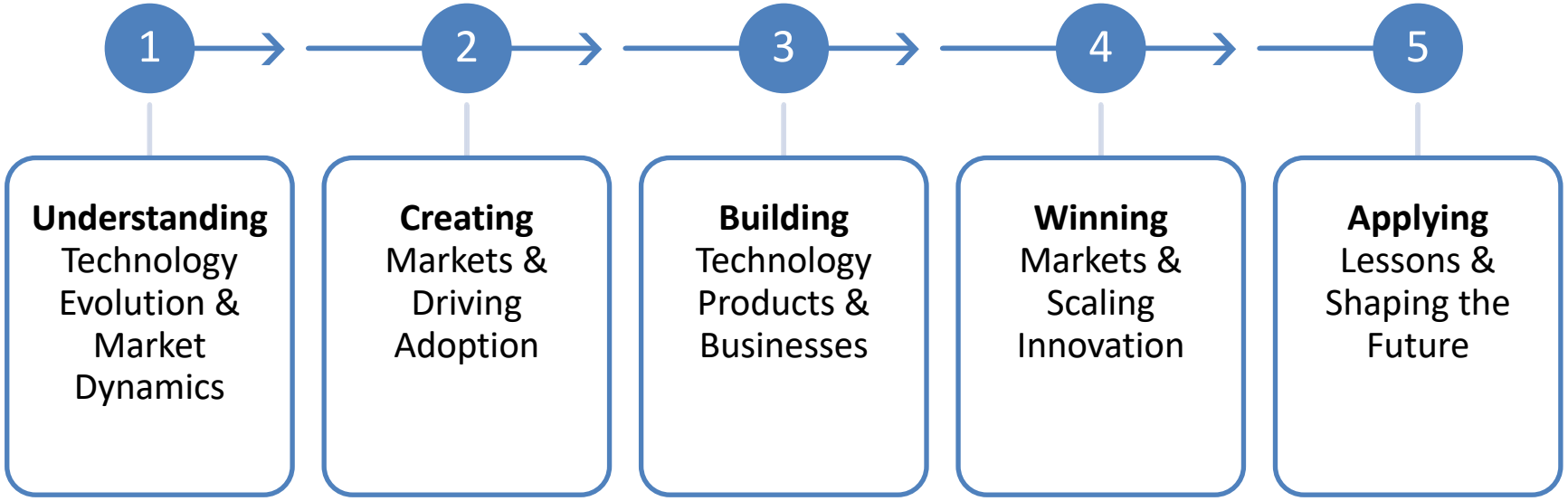
Participants are encouraged to actively contribute examples, articles, and perspectives from their own professional experiences, creating a collaborative learning environment that extends beyond the formal classroom discussions.

Evaluation

The students will be graded as follows on each of the segments for the final evaluation:

Technology Business Project (Group):	25%
Case Analysis (Group):	1 x 15 = 15%
Case Analysis (Individual.):	3 x 10 = 30%
Simulation & Class Room Engagement:	15%
Class Contribution & Participation:	15%

Learning Journey



SMIT develops an integrated perspective spanning technology evolution, market development, and industry transformation. Rather than teaching individual technologies, it equips students with enduring frameworks for creating, commercializing, and scaling technology businesses across successive waves of technological change.

Session Plan (Indicative)

Block I: Understanding Technology Evolution & Market Dynamics			
Sl.	Title	Key Themes & Outcomes	Class Activity & Required Readings (prior to class)
1	**Introduction to Technology Markets**	• Innovation vs. invention • Technology market characteristics • Sources of competitive advantage • Marketing's evolving role	**Architectural Innovation: The reconfiguration of existing product technologies etc** by *Henderson & Clark*
2	**Technology Evolution & Industry Dynamics**	• Technology evolution patterns • Industry disruption • Standards & compatibility • Technology migration	Class Discussion: T**he Disruption of Indian Telecom (B) - From Voice to Data** by *Bagri* **Performance Provided, Market Demand, & the Product Life Cycle** by *Christensen*
3	**Competing by Redefining Markets**	• Driving transitions • Changing basis of competition • Network driven markets • Managing complementors	Case * : **Responding to the Wii?** by *Hagiu & Halaburda* **Darwin & the Demon: Innovating Within Established Enterprises** by *Moore*
4	**Digital Transformation & Industry Dynamics**	• Digitization and digitalization • Modularity implications • Nature of digital disruption • Value migration	Case*: **AmTran Technology** Ltd. by *Shih, Wang & Robinson* **Managing in an Age of Modularity** by *Baldwin & Clark* **How to Anticipate Value Migration** by *Slwyotzky*

Block II: Creating Markets and Driving Adoption			
Sl.	Title	Key Themes & Outcomes	Class Activity & Required Readings (prior to class)
5	**Understanding Technology Adoption**	• Diffusion of innovation • Technology adoption lifecycle • Crossing the chasm • Whole product concept	Simulation: **Innovation Marketing Simulation: Crossing the Chasm**, by *Eckhardt, Cavender, Moore & Gill*
6	**Designing Successful Technology Products and Platforms**	• Designing breakthrough products • Identifying the beachhead segment • Developing the whole product • Platform strategies	Class*: **Linden Lab: Crossing the Chasm** by *Eisenmann & Wagonfeld* **Staged Solution to the Catch-22** by *Hagiu & Eisenmann*
7	**Creating Markets for New Technologies**	• Bringing new technology to market • Defining the competitive frame • Managing the risk • Integrated marketing management	Case **: **NTT DoCoMo: Marketing i-mode** by Moon **Versioning: The Smart Way to Sell Information** by *Shapiro & Varian* **Ecosystem Advantage: How to successfully etc.** by *Williamson & De Meyer*
8	**Driving Industry Disruption**	• Deliberate vs emergent strategy • Value network redesign • Business model innovation • Dominant design & industry standards	Case**: **Netflix** by *Shih, Kaufman & Spinola* **Art of Standards Wars** by *Shapiro & Varian* **Reinventing your Business Model** by *Johnson, Christensen & Kagermann*

Block III: Building Technology Products & Businesses			
Sl.	Title	Key Themes & Outcomes	Class Activity & Required Readings (prior to class)
9	**Technology Strategy & Product Management**	• Technology strategy • Product platform strategy • Product portfolio decisions • Technology roadmaps	Case*: **The Disruption of Indian Telecom (A) - The Launch of Reliance Jio** by *Bagri*
10	**Commercialising Frontier Technologies**	• Designing breakthrough products • Customer-centred design • Product-market fit • Product and service platforms	Case**: **Copilot(s): Generative AI at Microsoft & GitHub** by *Nagle, Greenstein, Roche, Wright & Mehta*
11	**Competing Through Business Model Innovation**	• Commercialisation strategy • Pricing & monetisation • AI enabled business models • Product launch strategy	Case*: **Pinduoduo: 300+M Shoppers Teaming for Good Deals** by *Zhou & Lau* **Ecommerce Disrupted: Is Social Ecommerce the Future of Online Shopping?** by *Sinha, Bagri, Pedada & Srivastava* **The Procrustean Bed Revisited: Ways to Scale 'Physical Service' Platforms** by *Bagri*
12	**Building Partnerships, Platforms & Ecosystems**	• Nature of strategic partnerships • Partnering across the life cycle • Product management • Ecosystem orchestration	**Match Your Innovation Strategy to Your Innovation Ecosystem** by *Adner* **Right Tech, Wrong Time** by *Adner & Kapoor*

Block IV: Winning Markets and Scaling Innovation			
Sl.	Title	Key Themes & Outcomes	Class Activity & Required Readings (prior to class)
13	**Scaling Technology Businesses**	• Category creation • Scaling innovation • Brand leadership • Growth through ecosystems	Case*: **Tesla Inc.** by *Rothaermel & King*
14	**Marketing Technology Products**	• Go-to-market strategy • Technology pricing • Customer success • Product-led growth	**Product-Led Growth Companies Find a New Way to Serve Customers** by *Libert & Davenport* **How Software Companies Can Avoid the Trap of Product-Led Growth** by. *Bussgang*
15	**Managing Technology Across Industries**	• Cross-industry applications • Technology convergence • Adjacent market expansion • Strategic opportunity mapping	**Competing in the Age of Digital Convergence** by *Yoffie*
16	**From Products to Solutions**	• Product, service & solution strategies • Cross-selling & solution selling • Enterprise relationship management • Outcome-based value propositions	In-class Case: **Cross Selling or Cross Purposes?** by *Harding*

Block V: Applying Lessons and Shaping the Future			
Sl.	Title	Key Themes & Outcomes	Class Activity & Required Readings (prior to class)
17 18	**Student Project Presentations**	• Industry diagnosis • Opportunity assessment • Strategic recommendations • Business model evaluation • Competitive positioning • Growth strategy • Peer learning	**Group Presentations**
19	**Technology Frontiers & Future Competition**	• AI agents & robotics • Quantum possibilities • Future business models • Strategic implications	**AI Agents Aren't Ready for Consumer-Facing Work** by *Furr, Gaarlandt, Mohan & Shipilov* **The Business Case for Quantum Computing** by *Bova, Goldfarb & Melko*
20	**Building the Next Technology Business**	• Strategic integration • Technology venture blueprint • Future competitive advantage • Leadership lessons	

About the Instructor



Dr. Prakash Bagri

Professor, Innovation & Marketing Strategy
Founder & Principal, PRB Partners

Dr. Prakash Bagri is a seasoned academic and industry expert with over three decades of experience in **innovation, technology commercialization, and business transformation**. As the Founder and Principal of **PRB Partners**, a boutique consulting firm, he helps organizations navigate strategic change, innovation, and business transformation.

Dr. Bagri has held senior academic roles at the **Indian School of Business** and served as adjunct faculty at the **Indian Institute of Management** in Ahmedabad, Bangalore, and Calcutta. In his corporate career, he held senior leadership positions at **Intel** and Unilever, where he built billion-dollar businesses, pioneered new product categories, and led transformative initiatives that significantly advanced technology adoption in India.

His research focuses on innovation in the Indian context, digital platforms, and value growth in Global Capability Centres (GCCs), with an emphasis on inclusive innovation and resilient business models. He has published papers on hybrid platforms and social e-commerce and is currently exploring the impact of digital public goods on India's soft power globally.

Dr. Bagri has provided strategic advisory, mentorship, and executive education to a diverse range of global and Indian organizations, including **Reliance Industries, Tata Group, Merck, Microsoft, KPMG, Accenture, Deloitte, Fidelity, Barclays, HDFC Life, State Bank of India, TVS Group, Bosch India, Cipla, Merck, Royal Enfield**, and many others.

He holds a doctorate from **Singapore Management University**, an MBA from **IIM Bangalore**, and a degree from **IIT Kharagpur**.

For further details, click here.

Group Technology Project

The **Group Technology Project** provides participants with an opportunity to integrate and apply the concepts developed throughout the course to a contemporary business challenge. Working in teams, participants will select an existing industry and examine how an emerging technology, or a novel application of an established technology, can create new market opportunities, reshape competitive dynamics, and generate sustainable business value.

Rather than preparing a conventional industry analysis, each team is expected to develop a technology enabled business opportunity supported by a coherent market strategy. The emphasis is on applying the frameworks discussed in the course to develop an integrated perspective spanning technology evolution, market opportunities, customer adoption, business model design, partnerships, commercialization, and industry impact.

Projects are expected to demonstrate rigorous analysis, originality of thought, practical feasibility, and strong managerial relevance. Particular emphasis will be placed on the ability to connect technological possibilities with customer needs and market realities, while developing a compelling pathway for value creation, delivery, and long-term competitive advantage.

Teams will present their work during the final sessions of the course, providing an opportunity for peer learning, discussion, and feedback. Detailed project guidelines, evaluation criteria, and suggested topic areas will be shared separately during the course.